

Hellenic Complex Systems Laboratory

# Inferences about the Difference between Two Proportions

Technical Report XVIII

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2018

# Inferences about the Difference between Two Proportions

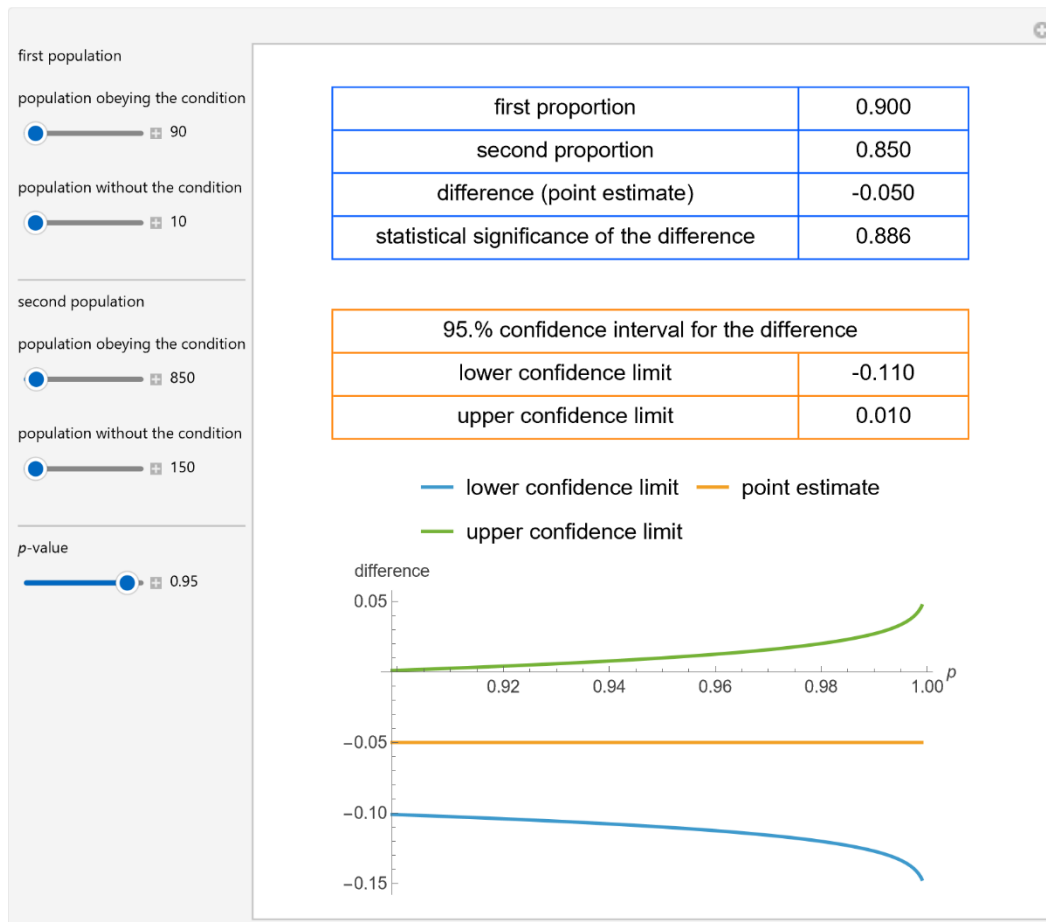
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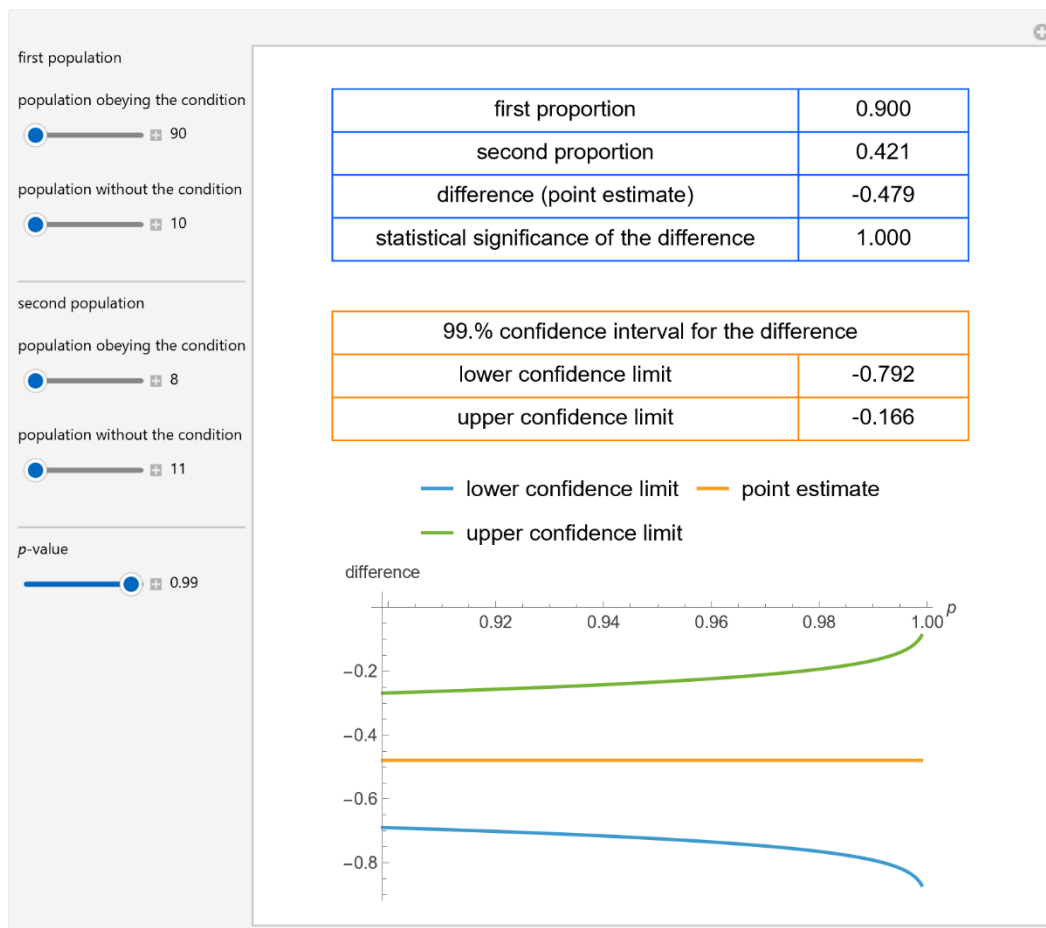
**Search Terms:** proportions, difference between proportions, confidence interval, inference, critical ratio statistic, statistical significance, statistics

## Short Description of the Demonstration

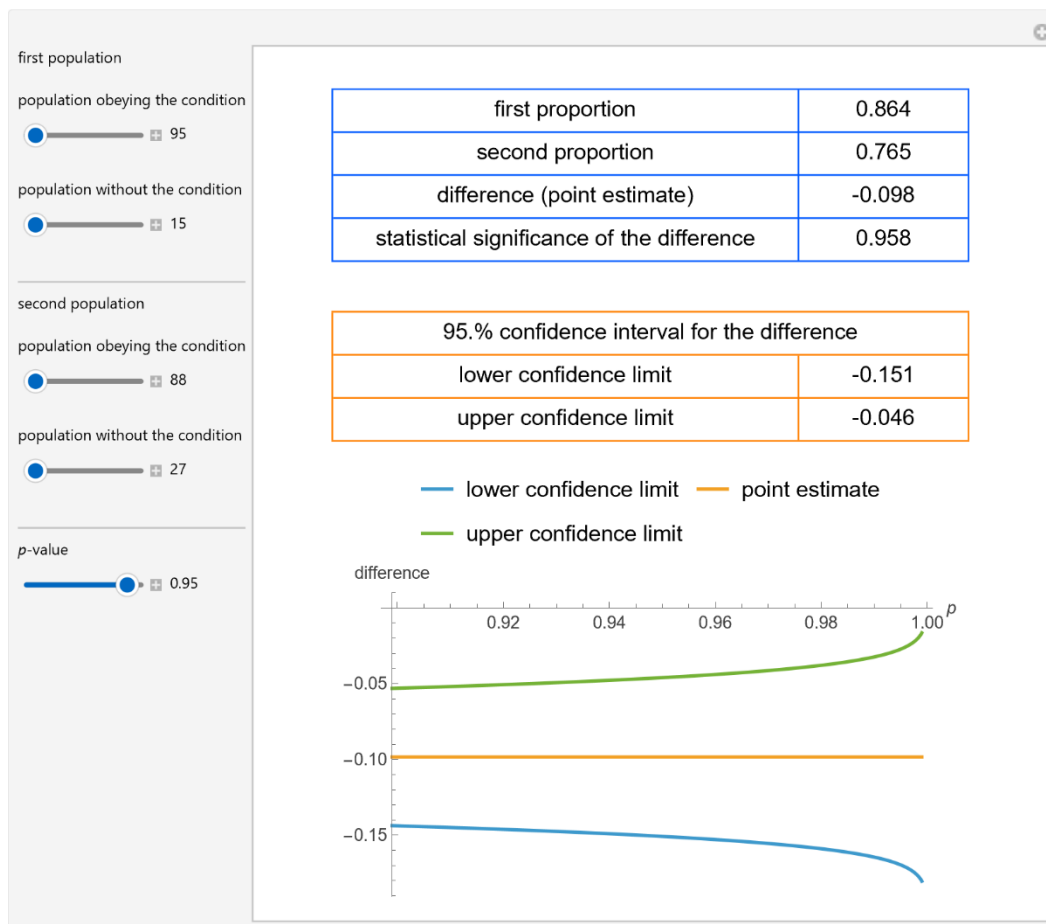
This Demonstration explores statistical inferences regarding the difference between two proportions of populations with a condition (or trait). It calculates the statistical significance and confidence intervals for the difference between two proportions and plots the confidence limits as functions of the selected confidence level. This is done for different population counts with and without the condition, and for different confidence levels used in the estimation of the confidence intervals.



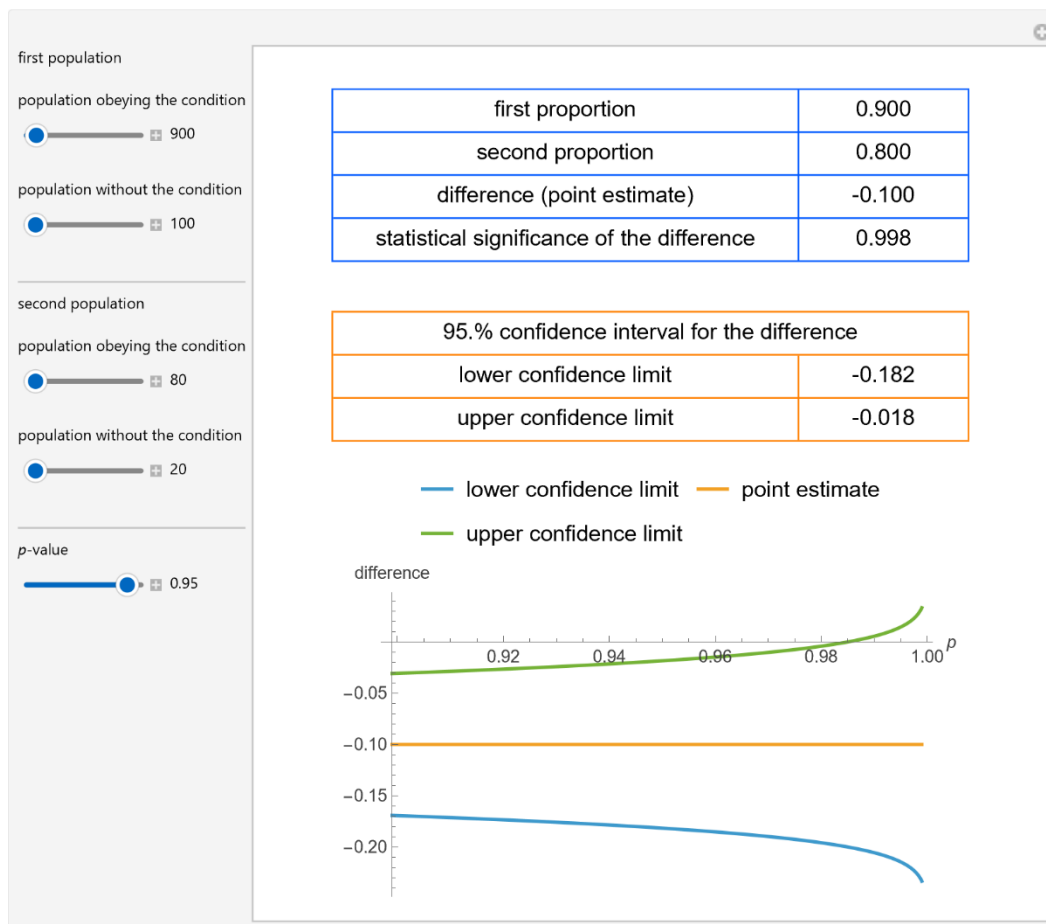
**Figure 1:** Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.



**Figure 2:** Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.



**Figure 3:** Point estimate and 99% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.



**Figure 4:** Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus confidence level for the difference, with the settings shown at the left.

## Details

The statistical significance of the difference between the two proportions is assessed by means of the z-score (or critical ratio statistic) [1]. The confidence interval for the difference between the proportions is calculated as in [2].

## References

[1] J. L. Fleiss, B. Levin and M. C. Paik. Methods for Generating a Fourfold Table. *Statistical Methods for Rates and Proportions*, 3rd ed., Hoboken, NJ: J. Wiley, 2003 pp. 51–52.

[2] J. L. Fleiss, B. Levin and M. C. Paik. A Simple Confidence Interval for the Difference between Two Independent Proportions. *Statistical Methods for Rates and Proportions*, 3rd ed., Hoboken, NJ: J. Wiley, 2003 pp. 60–61.

## Demonstration

### Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/InferencesAboutTheDifferenceBetweenTwoProportions.nb>

### Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

### System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

### Permanent Citation:

Chatzimichail T, Hatjimihail AT. Inferences about the Difference between Two Proportions. Wolfram Demonstrations Project, Champaign: Wolfram Research, Inc., 2018. Available at:

<https://demonstrations.wolfram.com/InferencesAboutTheDifferenceBetweenTwoProportions/>

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## Technical Report

### DOI

[10.5281/zenodo.20291340](https://doi.org/10.5281/zenodo.20291340)

### Citation

Chatzimichail T, Hatjimihail AT. *Inferences about the Difference between Two Proportions*. Technical Report XVIII. Hellenic Complex Systems Laboratory; 2018. <https://doi.org/10.5281/zenodo.20291340>

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First Published: August 2, 2018

Revised: May 19, 2026